

Verification Continuum™

Verdi® FSDB Writer Summary

Version W-2024.09, September 2024



Copyright and Proprietary Information Notice

© 2024 Synopsys, Inc. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>. All other product or company names may be trademarks of their respective owners.

Free and Open-Source Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

www.synopsys.com

Contents

Customer Support	4
Synopsys Statement on Inclusivity and Diversity	4

1. Verdi FSDB Writer Summary	6
Verdi 2023.12	6
Verdi 2017.12	6
Verdi 2016.06	6
Verdi 2015.09	7
Verdi 2012.10	7
Verdi 2012.01	7

Preface

The Verdi FSDB Writer Summary provides detailed information of the FSDB Writer updates which occurred in different Verdi versions.

Customer Support

For any online access to the self-help resources, you can refer to the documentation and searchable knowledge base available in SolvNetPlus.

To obtain support for your Verdi product, choose one of the following:

- Open a case through SolvNetPlus.

Go to <https://solvnetplus.synopsys.com/s/contactsupport> and provide the requested information, including:

- Product L1 as Verdi
- Case Type

Fill in the remaining fields according to your environment and issue.

- Send an e-mail message to verdi_support@synopsys.com.

Include product name (L1), sub-product name/technology (L2), and product version in your e-mail, so it can be routed correctly.

Your e-mail will be acknowledged by automatic reply and assigned a Case number along with Case reference ID in the subject (ref:____:ref).

For any further communication on this Case via e-mail, send e-mail to verdi_support@synopsys.com and ensure to have the same Case ref ID in the subject header or else it will open duplicate cases.

Note:

In general, we need to be able to reproduce the problem in order to fix it, so a simple model demonstrating the error is the most effective way for us to identify the bug. If that is not possible, then provide a detailed explanation of the problem along with complete error and corresponding code, if any/permissible.

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also

Contents

includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Verdi FSDB Writer Summary

This chapter covers the previous release history for the FSDB writer.

Verdi 2023.12

Starting from this release, you can migrate to NPI Waveform Model/Writer, which is the latest FSDB Reader/Writer to seamlessly read FSDB. The following data types are supported only with NPI Waveform Model/Writer:

1. Modport/interface port
 2. Virtual file
 3. The FSDB which includes power signals
 4. Packed mode dumping
-

Verdi 2017.12

- Starting from this release, the FSDB is upgraded from version 5.5 to version 5.6.
 - The `fsdbVarIdcode` is enlarged from `int` to `int64` so that larger designs (over 2G idcode) can be handled with ease in the future.
-

Verdi 2016.06

Starting from this release, functions and tasks can be dumped to FSDB. By default, they are not dumped for performance consideration. You can use one of following methods to dump them to FSDB during runtime (simulation)

1. Using the `FSDB_FUNCTIONS` environment variable:

For example: `setenv FSDB_FUNCTIONS 1`

2. Using the `+fsdb+functions` runtime option:

For example: `simv +fsdb+functions`

3. Adding the `+functions` option to your `$fsdbDumpvars` dumping task:

For example: `$fsdbDumpvars "+functions")`

4. Adding the `+functions` option to your `fsdbDumpvars` Tcl command:

For example: `fsdbDumpvars +functions`

Verdi 2015.09

Starting from this release, the support for FSDB API static library(.a) is stopped.

Verdi 2012.10

Add `ffw_GetFsdbDumperVersion` to get the FSDB dumper version.

Verdi 2012.01

The Verdi version includes a newly-engineered FSDB that has been optimized for significant size savings. The version of this new FSDB file format is FSDB 5.0.

FSDB files generated by the Verdi 2012.01 FSDB dumper or writer can only be loaded into Verdi 2012.01 or later versions.

- The FSDB 5.0 version for the Verdi Reader has exactly the same API interface as the FSDB 4.x (Novas 2011.10) version. Header files are not changed.
- The Verdi Writer can create the FSDB 5.0 format, but the FSDB 5.0 file cannot be read by the FSDB Reader in versions earlier than FSDB 5.0.

To dump the FSDB 4.3 file, use the `NOVAS_FFW_ENV_OLD_FSDB` environment variable:

```
setenv NOVAS_FFW_ENV_OLD_FSDB 1
```